

Student Number: _____

Class Teacher: _____

St George Girls High School

Trial Higher School Certificate Examination

2019



Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using black pen.
- Board-approved calculators may be used.
- A reference sheet is provided.
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for incomplete or poorly presented solutions.

Section I	/10
Section II	
Question 11	/10
Question 12	/10
Question 13	/10
Question 14	/10
Question 15	/10
Question 16	/10
Total	/70

Total Marks – 70

Section 1

Pages 3 – 6

10 marks

- Attempt Questions 1 – 10.
- Allow about 15 minutes for this section.
- Answer on the multiple choice answer sheet provided at the back of this paper.

Section II

Pages 7 – 13

60 marks

- Attempt Questions 11 – 16.
- Allow about 1 hour and 45 minutes for this section.
- Begin each question in a new writing booklet.

Section I

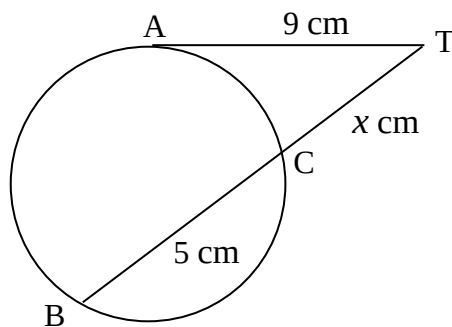
10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 - 10

1. The line AT is the tangent to the circle at A and the line BT is a secant meeting the circle at B and C, as shown in the diagram.



Given that $AT = 9$, $BC = 5$ and $CT = x$, which one of the following equations is correct?

- (A) $x^2 + 5x - 81 = 0$
(B) $x^2 + 5x + 81 = 0$
(C) $x^2 - 5x - 81 = 0$
(D) $x^2 + 5x - 9 = 0$
2. The acute angle between the lines $y = 2x + 4$ and $5x - y + 34 = 0$, to the nearest degree is:
- (A) 4°
(B) 7°
(C) 15°
(D) 74°

3. If $t = \tan \frac{\theta}{2}$, what is the correct expression for $\frac{1 - \cos \theta}{\sin \theta}$.

(A) $\frac{1}{t}$

(B) t

(C) $2t$

(D) $\frac{2}{t}$

4. Find the Cartesian equation of the curve defined by the parametric equations:

$$\begin{aligned}x &= \sin \theta \\y &= \cos^2 \theta - 3.\end{aligned}$$

(A) $y = -3 + 3x^2$

(B) $y = \sin^2 x - 3$

(C) $y = -2 - x^2$

(D) $y = \sin 2x + 3 \cos^2 x$

5. What is the value of $\lim_{x \rightarrow 0} \frac{\sin 2x}{x \cos x}$?

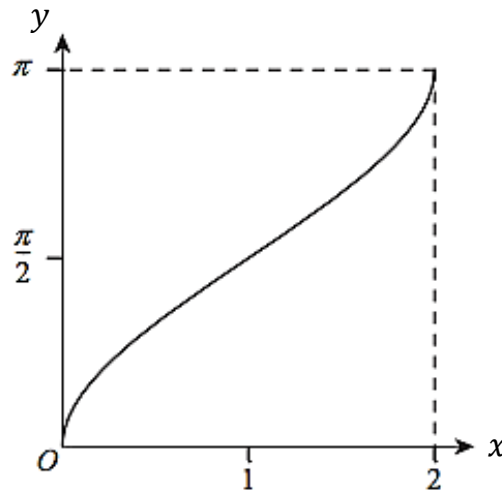
(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) 2

6. Consider the graph below.



Which function best describes this graph?

- (A) $y = \cos^{-1}(x)$
(B) $y = 1 - \cos^{-1}(x)$
(C) $y = \cos^{-1}(x - 1)$
(D) $y = \cos^{-1}(1 - x)$
7. $\int 4 \cos^2 4x \, dx =$
- (A) $\left(2x + \frac{1}{4} \sin 8x\right) + C$
(B) $\left(x + \frac{1}{2} \sin 8x\right) + C$
(C) $\left(x + \frac{1}{2} \cos 8x\right) + C$
(D) $\left(x + \frac{1}{4} \sin 8x\right) + C$

8. The velocity of a particle is given by $v = \sqrt{2 - x}$, where x is its displacement in metres and velocity (m/s).
Which of the following is a correct expression for the acceleration $\ddot{x}$?
- (A) $\ddot{x} = \frac{1}{2} \text{ m/s}^2$
(B) $\ddot{x} = \frac{1}{4} \text{ m/s}^2$
(C) $\ddot{x} = -\frac{1}{2} \text{ m/s}^2$
(D) $\ddot{x} = -\frac{1}{4} \text{ m/s}^2$
9. Which of the following is a general solution of the equation $\sin \frac{x}{2} = \sin \frac{\pi}{10}$?
- (A) $x = n\pi + (-1)^n \frac{\pi}{5}$
(B) $x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{20}$
(C) $x = 2n\pi + (-1)^n \frac{\pi}{10}$
(D) $x = 2n\pi + (-1)^n \frac{\pi}{5}$
10. The cubic curve $y = x^3 + 3ax + b$ has two turning points and crosses the y – axis at $(0, -a)$.
Which of the following could be true?
- (A) $a < 0$ and $b > 0$
(B) $a > 0$ and $b < 0$
(C) $a > 0$ and $b > 0$
(D) $a < 0$ and $b < 0$

End of Section I

Section II

60 marks

Attempt Questions 11 – 16

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11	(10 marks) Use a separate writing booklet	Marks
(a)	The point $C(x, y)$ divides the interval joining $A(-4, 8)$ to $B(6, -12)$ internally in the ratio $2 : 3$. Find the coordinates of C .	2
(b)	Solve for x : $\frac{x}{1-3x} \geq 1$.	3
(c)	Show that $\frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x} = 2$.	2
(d)	Find the exact value of $\int_0^{\frac{\sqrt{5}}{2}} \frac{dx}{\sqrt{5-4x^2}}$.	3

Question 12 (10 marks) Use a separate writing booklet

Marks

- (a) If α, β and γ are the roots of the equation $2x^3 + 5x - 3 = 0$, find the value of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$. 2
- (b) The rate of increase of a population P of sandflies on the track to Culbarra Beach is proportional to the difference between the population, P , and 2000. This rate is expressed by the differential equation $\frac{dP}{dt} = k(P - 2000)$, where k is a constant and t represents time in weeks.
- (i) Show that $P = 2000 + Ae^{kt}$, where A is a constant, satisfies the differential equation 1
- (ii) Initially, the population was 2500 and two weeks later it had increased to 5000. Find the value of A and k . 2
- (iii) After how many weeks will the population of sandflies exceed 10 000? 2
- (c) Evaluate $\int_{-1}^2 x\sqrt{3-x} \, dx$ using the substitution $u = 3 - x$, $x < 3$. 3

Question 13 (10 marks) Use a separate writing booklet

Marks

(a) Consider the continuous function $f(x) = \sec^{-1}x + \sin^{-1}\frac{1}{x}$ for $x \geq 1$.

(i) Show that $\sec^{-1}x = \cos^{-1}\frac{1}{x}$.

1

(ii) Hence, show that $f'(x) = 0$.

2

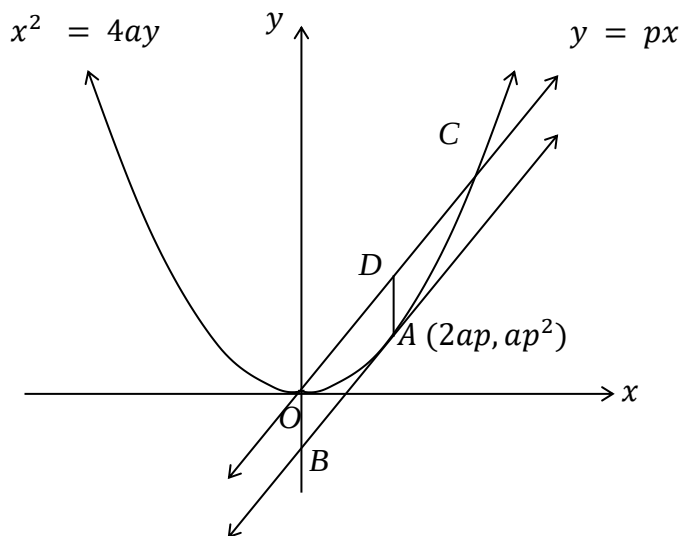
(iii) Hence, or otherwise, prove that

$$\sec^{-1}x + \sin^{-1}\frac{1}{x} = \frac{\pi}{2} \text{ for } x \geq 1.$$

2

(b) The point $A(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$.
 The tangent at A intersects the y -axis at B .

The line $y = px$ crosses the parabola at the origin and at the point C .
 Let D be the midpoint of OC .



(i) Show that the coordinates of D is $(2ap, 2ap^2)$.

1

(ii) Show that the coordinates of B is $(0, -ap^2)$.

2

(iii) Show that $ODAB$ is a parallelogram.

2

Question 14 (10 marks) Use a separate writing booklet

Marks

(a) Consider the function $f(x) = 3 \sin 2x - x$.

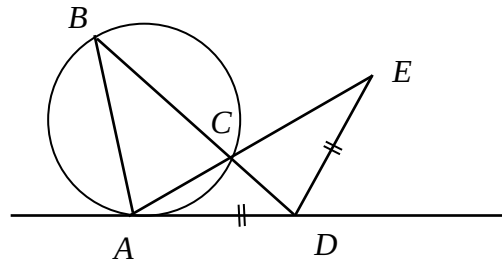
(i) Show that $f(x) = 0$ has a root α such that $1.33 < \alpha < 1.34$.

2

(ii) Starting with $\alpha = 1.33$, use one application of Newton's Method to find a better approximation for this root, giving your answer to 4 decimal places.

2

(b)



In the diagram, ABC is a triangle inscribed in the circle. The tangent to the circle at A meets BC produced to D . E is the point on AC produced such that $DA = DE$.

3

Copy the diagram into your booklet.

Prove that $ABED$ is a cyclic quadrilateral.

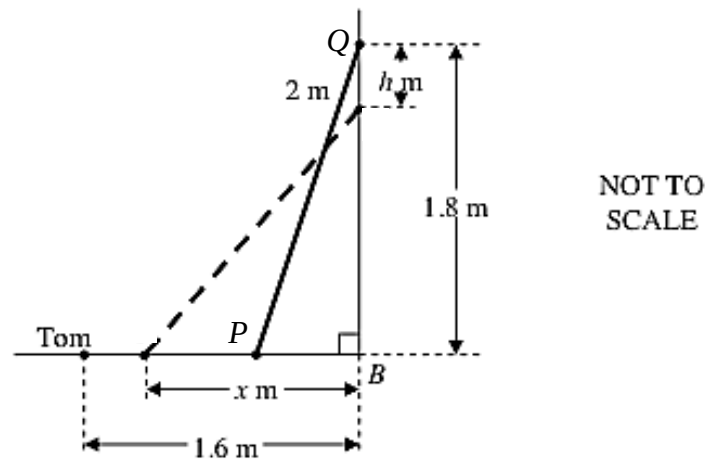
(c) Use mathematical induction to prove that $4^n + 6n - 1$ is divisible by 9 for integers $n \geq 1$.

3

Question 15 (10 marks) Use a separate writing booklet

Marks

- (a) A particle is moving so that its distance x centimetres from a fixed point O at time t seconds is $x = 6 \sin 2t$.
- (i) Show that the particle is moving in simple harmonic motion. 2
- (ii) Find the period of the motion. 1
- (iii) Find the velocity of the particle when it first reaches 3 centimetres to the right of the origin. 2
- (b) The diagram shows a ladder PQ , 2 metres in length, leaning against a wall such that the top of the ladder, Q , initially reaches 1.8 metres up the wall. The base of the ladder, P , is x metres from the base of the wall, B .



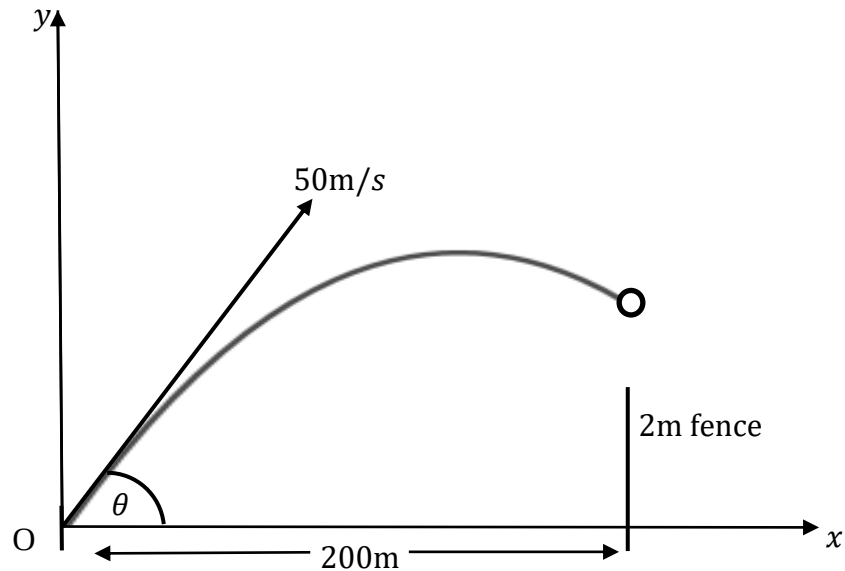
The ladder begins to slide down the wall at the rate of 0.5 metres per minute such that the top of the ladder is h metres below its original position after t minutes.

- (i) Show that t minutes after the ladder begins to slide down the wall,
- $$h = 1.8 - \sqrt{4 - x^2}. \quad 2$$
- (ii) Tom is standing on the ground 1.6 metres from the base of the wall in a direct line with the ladder. At what rate does base of the ladder hit Tom? 3

Question 16 (10 marks) Use a separate writing booklet

Marks

- (a) A method to score a home run in a baseball game is to hit the ball over the boundary fence on the full.



A ball is hit at 50 m/s. The fence, 200 metres away, is 2 metres high. You may neglect air resistance and acceleration due to gravity can be taken as 10 m/s^2 . You may assume the following equations of motion:

$$x = 50t \cos \theta \quad \text{and} \quad y = 50t \sin \theta - 5t^2$$

**DO NOT PROVE
 THESE EQUATIONS**

- (i) Show that the Cartesian equation of motion is given by 2

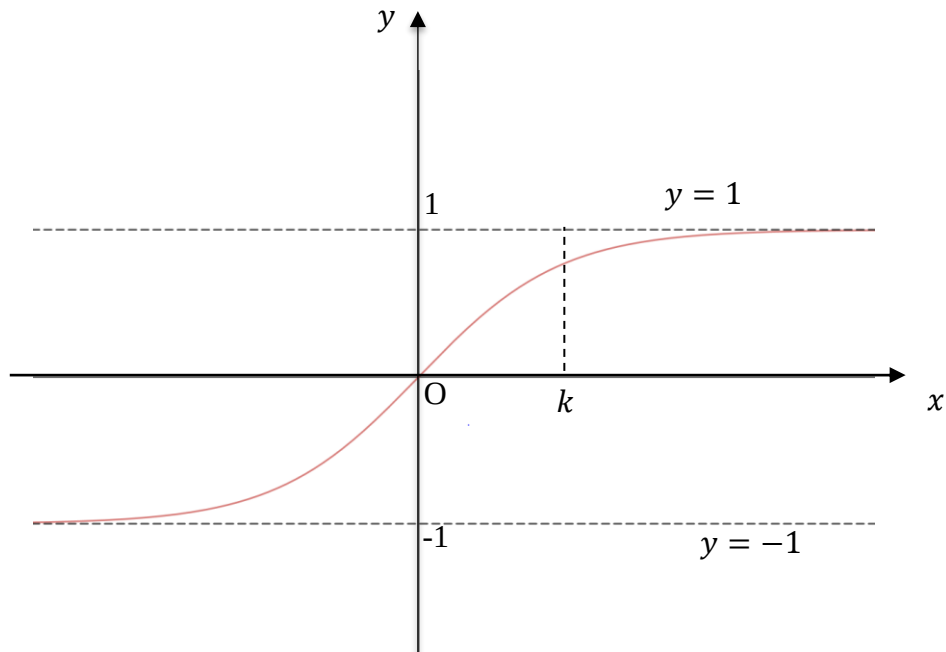
$$y = x \tan \theta - \frac{x^2}{500} (\sec^2 \theta), \quad \text{where } \theta \text{ is the angle of projection.}$$

- (ii) Show that if the ball just clears the 2 metre boundary fence then, 2

$$40 \tan^2 \theta - 100 \tan \theta + 41 = 0.$$

- (iii) In what range of values must θ lie to score a home run by this method? 2

(b)



The curve shows the graph of the function

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

where $y = \pm 1$ are the horizontal asymptotes.

- (i) If k is a positive constant, show that the area in the first quadrant enclosed by the above curve, the lines $y = 1$, $x = 0$ and $x = k$ is given by:

$$A = k - \ln(e^k + e^{-k}) + \ln 2$$

2

- (ii) By considering the area in (i), prove that for all positive values of k , the area is always less than $\ln 2$.

2

END OF PAPER

MATHEMATICS EXTENSION 1 - QUESTION Multiple Choice

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
1. $9^2 = x(x+5)$ $81 = x^2 + 5x$ $x^2 + 5x - 81 = 0$	A	
2. For $y = 2x + 4$, $m_1 = 2$ $5x - y + 34 = 0$, $m_2 = 5$ $\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right$ $= \left \frac{2 - 5}{1 + 2(5)} \right$ $= \frac{3}{11}$ $\theta \approx 15^\circ$	C	
3. $\frac{1 - \cos \theta}{\sin \theta} = \frac{1 - \frac{1 - t^2}{1 + t^2}}{\frac{2t}{1 + t^2}} \times 1 + t^2$ $= \frac{1 + t^2 - (1 - t^2)}{2t} \times 1 + t^2$ $= \frac{1 + t^2 - 1 + t^2}{2t}$ $= \frac{2t^2}{2t}$ $= 1$	B	

MATHEMATICS EXTENSION 1 – QUESTION MC

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

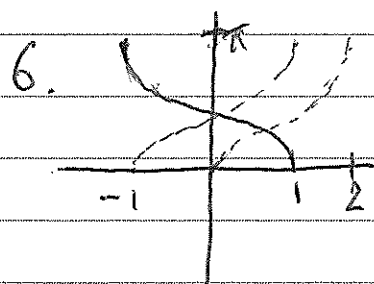
4. $x = \sin \theta$ — — — (1)
 $y = \cos^2 \theta - 3$ — — — (2)

From (2) $y = (1 - \sin^2 \theta) - 3$
 $= -\sin^2 \theta - 2$
 $= -x^2 - 2$ from (1)

C

5. $\lim_{x \rightarrow 0} \frac{\sin 2x}{x \cos x} = \lim_{x \rightarrow 0} \frac{2 \sin x \cos x}{x \cos x}$
 $= 2 \lim_{x \rightarrow 0} \frac{\sin x}{x}$
 $= 2 \times 1$
 $= 2$

D



$y = \cos^{-1} x$
 $y = \cos^{-1}(-x)$
 $y = \cos^{-1}(-x+1)$
 $y = \cos^{-1}(1-x)$

D

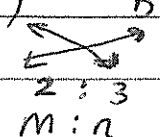
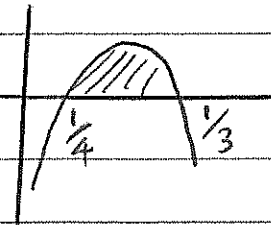
7. $\int 4 \cos^2 4x \, dx$
 $= 4 \int \cos^2 4x \, dx$
 $= 4 \int \frac{1}{2} (\cos 8x + 1) \, dx$
 $= 2 \left[\frac{\sin 8x}{8} + x \right] + c$
 $= 2x + \frac{1}{4} \sin 8x + c$

A

MATHEMATICS EXTENSION 1 – QUESTION MC

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
8. $v = \sqrt{2-x}$ $v^2 = 2-x$ $\dot{x} = \frac{d\left(\frac{1}{2}v^2\right)}{dx}$ $= \frac{d\left(\frac{1}{2}(2-x)\right)}{dx}$ $= \frac{d\left(1 - \frac{x}{2}\right)}{dx}$ $= -\frac{1}{2} \text{ m/s}^2$	C	
9. $\sin \frac{x}{2} = \sin \frac{\pi}{10}$ $\frac{x}{2} = \sin^{-1}\left(\sin \frac{\pi}{10}\right)$ $\frac{x}{2} = \frac{\pi}{10}$ $\frac{x}{2} = n\pi + (-1)^n \frac{\pi}{10}$ $\frac{x}{2} = n\pi + (-1)^n \frac{\pi}{10}$ $x = 2n\pi + (-1)^n \frac{\pi}{5}$	D	
10. $y' = 3x^2 + 3a$ For $\pm p$, $y' = 0$ $3x^2 + 3a = 0$ $x^2 + a = 0$ Two real solutions if $a < 0$. As curve passes through $(0, -a)$ then $y = -a$ and $x = 0$. So $-a = 0 + b$ $b = -a$ So $a < 0$ then $b > 0$	A	

MATHEMATICS EXTENSION I – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) $A(-4, 8) \quad B(6, -12)$  $x = \frac{nx_1 + mx_2}{m+n}$ $= \frac{3(-4) + 2(6)}{2+3}$ $= \frac{-12 + 12}{5}$ $= 0$ $y = \frac{ny_1 + mx_2}{m+n}$ $= \frac{3(8) + 2(-12)}{2+3}$ $= \frac{24 - 24}{5}$ $= 0$ $\therefore C$ is $(0, 0)$		2 marks for both values of x and y i.e $C(0, 0)$ 1 mark for either x or y correctly found.
b) $\frac{x}{1-3x} \geq 1, \quad x \neq \frac{1}{3}$ Multiplying both sides by $(1-3x)^2$ $\frac{(1-3x)^2 \cdot x}{(1-3x)} \geq 1 \times (1-3x)^2$ $(1-3x)x \geq (1-3x)^2$ $x(1-3x) - (1-3x) \geq 0$ $(1-3x)[x - (1-3x)] \geq 0$ $(1-3x)(4x-1) \geq 0 \quad \dots (*)$  $\therefore$ $\frac{1}{4} \leq x < \frac{1}{3}$	1 1 1	Many students had problems drawing the quadratic. Either the concavity was incorrect or the x-intercepts were not labelled correctly.

MATHEMATICS EXTENSION I – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
Furthermore for Q11(b)		
• some students included $\frac{1}{3}$ in their solution and gave		
$\frac{1}{4} \leq x \leq \frac{1}{3} \Rightarrow$		$2\frac{1}{2}$ marks were awarded
• Many students used the incorrect sign to the quadratic inequality and therefore the solution obtained was		
$x \leq \frac{1}{4}$ or $x > \frac{1}{3}$		2 marks were awarded
11(c)		
$\text{LHS} = \frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x}$		
$= \frac{\cos x \sin 3x - \sin x \cos 3x}{\sin x \cos x}$	$\frac{1}{2}$	
$= \frac{\sin 3x \cos x - \cos 3x \sin x}{\sin x \cos x}$		1 mark
$= \frac{\sin(3x - x)}{\sin x \cos x}$		
$= \frac{\sin 2x}{\sin x \cos x}$		1 mark.
$= \frac{2 \sin x \cos x}{\sin x \cos x}$		
$= 2$		
$= \text{RHS}$		

MATHEMATICS EXTENSION I – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
NOTE:		
Many students thought that they could factorise,		
$\cos x \sin 3x - \sin x \cos 3x$ as $\cos x \sin x (\sin 2x - \cos x)$		
This is quite concerning and care needs to be taken.		
Please revise compound angles.		
Alternative solution (by many students) (but not preferred)		
LHS = $\frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x}$		
$= \frac{\sin(2x+x)}{\sin x} - \frac{\cos(2x+x)}{\cos x}$		
$= \frac{\sin 2x \cos x + \cos 2x \sin x}{\sin x} - \frac{\cos 2x \cos x - \sin 2x \sin x}{\cos x}$		1/2
$= \frac{\sin 2x \cos x}{\sin x} + \frac{\cos 2x \sin x}{\sin x} - \frac{\cos 2x \cos x}{\cos x} + \frac{\sin 2x \sin x}{\cos x}$		1/2
$= \frac{2 \sin x \cos x \cos x}{\sin x} + \frac{2 \sin x \cos x \sin x}{\cos x}$		1/2
$= 2 \cos^2 x + 2 \sin^2 x$		1/2
$= 2 (\cos^2 x + \sin^2 x)$		
$= 2$		
$= \text{RHS}$		

MATHEMATICS EXTENSION I – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$d) \int_0^{\frac{\sqrt{5}}{2}} \frac{dx}{\sqrt{5-4x^2}}$ $= \int_0^{\frac{\sqrt{5}}{2}} \frac{dx}{\sqrt{4\left(\frac{5}{4}-x^2\right)}}$ $= \frac{1}{2} \int_0^{\frac{\sqrt{5}}{2}} \frac{dx}{\sqrt{\frac{5}{4}-x^2}}$ $= \frac{1}{2} \int_0^{\frac{\sqrt{5}}{2}} \frac{dx}{\sqrt{\left(\frac{\sqrt{5}}{2}\right)^2-x^2}}$ $= \frac{1}{2} \left[\sin^{-1} \frac{x}{\frac{\sqrt{5}}{2}} \right]_0^{\frac{\sqrt{5}}{2}}$ $= \frac{1}{2} \left[\sin^{-1} \frac{2x}{\sqrt{5}} \right]_0^{\frac{\sqrt{5}}{2}}$ $= \frac{1}{2} \left[\sin^{-1} \frac{2 \cdot \frac{\sqrt{5}}{2}}{\sqrt{5}} - \sin^{-1} 0 \right]$ $= \frac{1}{2} \sin^{-1} 1$ $= \frac{1}{2} \times \frac{\pi}{2}$ $= \frac{\pi}{4}$	Many students took 2 out of the integrand instead of $\frac{1}{2}$. Many students took 2 out of the integrand instead of $\frac{1}{2}$. 1 $\frac{1}{2}$ $\frac{1}{2}$	Many students took 2 out of the integrand instead of $\frac{1}{2}$. $\therefore 2\frac{1}{2}$ marks awarded if answer was π. Similarly if nothing was taken out, $2\frac{1}{2}$ marks were awarded if the remaining solution was correct.
Note: Some students did not find the definite integral and just integrated to give $\frac{1}{2} \sin^{-1} \frac{2x}{\sqrt{5}} + c \Rightarrow$	1 mark awarded.	

MATHEMATICS EXTENSION 1 – QUESTION 12

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$a) 2x^3 + 5x - 3 = 0$$

$$a = 2$$

$$b = 0$$

$$c = 5$$

$$d = -3$$

$$\boxed{\alpha^{-1} + \beta^{-1} + \gamma^{-1} = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}}^*$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$$

$$= \frac{c/a}{-d/a}$$

$$= \frac{5/2}{3/2}$$

$$= \frac{5}{3}$$

* Many students missed this connection, and overcomplicated the question.

1

for either $5/2$
or $3/2$

1

Note: an answer of $-\frac{5}{3}$ was awarded $1\frac{1}{2}$ marks.

$$b) \text{ i } \text{ If } P = 2000 + Ae^{kt}$$

$$\text{then } \frac{dP}{dt} = kAe^{kt}$$

$$= k(2000 + Ae^{kt} - 2000)$$

$$= k(P - 2000)$$

so it satisfies the differential equation

} 1

Note: In a "show" question, you must show every step. Do not expect the examiner to join the dots for you. Many students left out crucial steps or substitutions, and only earned half marks.

MATHEMATICS EXTENSION 1 – QUESTION 12 (continued)

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

b) i - Alternative solution

* crucial step

$$\text{If } P = 2000 + Ae^{kt}, \text{ then } Ae^{kt} = P - 2000 \quad (*) \textcircled{1}$$

$$\frac{dP}{dt} = kAe^{kt}$$

$$= k(P - 2000) \quad (\text{from } \textcircled{1})$$

so it satisfies the differential equation

} 1 mark

ii when $t = 0$, $P = 2500$

$$\therefore 2500 = 2000 + Ae^{0k}$$

$$= 2000 + A$$

$$\therefore A = 500$$

1

when $t = 2$, $P = 5000$

$$\therefore 5000 = 2000 + 500e^{2k}$$

$$500e^{2k} = 3000$$

$$e^{2k} = 6$$

$$2k = \ln 6$$

$$k = \frac{1}{2} \ln 6$$

$$= 0.895879 \dots$$

$$\approx 0.896 \quad (3 \text{ d.p.})$$

1

iii Population will reach 10000 when

$$10000 = 2000 + 500e^{kt}$$

$$500e^{kt} = 8000$$

$$e^{kt} = 16$$

$$kt = \ln 16$$

1

MATHEMATICS EXTENSION 1 – QUESTION 12 (continued)

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$t = \frac{\ln 16}{k} \quad \text{but } k = \frac{1}{2} \ln 6$		
$= \frac{\ln 16}{\frac{1}{2} \ln 6}$		
$= \frac{2 \ln 16}{\ln 6}$		
$= 3.0948224 \dots$	1	
$\therefore$ the population will exceed 10000 during the 4th week.		
9) $\int_{-1}^2 x \sqrt{3-x} dx$		
$u = 3-x \Rightarrow x = 3-u$		
$\frac{du}{dx} = -1$		
$dx = -du$		
when $x = -1$, $u = 4$ when $x = 2$, $u = 1$	1	
$= \int_4^1 (3-u) \sqrt{u} (-du)$	1	Candidates who mistakenly made the integration easier could not earn the 3rd mark.
$= \int_1^4 3\sqrt{u} - u\sqrt{u} du$		
$= \left[2u^{\frac{3}{2}} - \frac{2u^{\frac{5}{2}}}{5} \right]_1^4$		
$= \frac{16}{5} - \frac{8}{5}$		
$= \frac{8}{5}$	1	

MATHEMATICS EXTENSION I – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a)(i) let $\alpha = \sec^{-1} x$ $\sec \alpha = x$ $\cos \alpha = \frac{1}{x}$ $\therefore \alpha = \cos^{-1} \frac{1}{x}$ $= \text{RHS}$ OR LHS = $\cos^{-1} \frac{1}{x}$ let $\alpha = \cos^{-1} \frac{1}{x}$ $\cos \alpha = \frac{1}{x}$ $\sec \alpha = x$ $\therefore \alpha = \sec^{-1} x$	(1)	Poorly attempted NOTE: $\sec^{-1} x \neq (\sec x)^{-1}$
(ii) $f(x) = \sec^{-1} x + \sin^{-1} \frac{1}{x}$ $= \cos^{-1} \frac{1}{x} + \sin^{-1} \frac{1}{x}$ from (i)		
Consider $\frac{d}{dx} (\sin^{-1} \frac{1}{x})$ let $u = \frac{1}{x}$ $= \frac{d}{dx} (\sin^{-1} u)$ $u = x^{-1}$ $= \frac{1}{\sqrt{1-u^2}}$ $\frac{du}{dx} = -x^{-2}$ $= -\frac{1}{x^2}$		
$\therefore \frac{dy}{dx} = \frac{1}{\sqrt{1-u^2}} \times -\frac{1}{x^2}$ $= \frac{-1}{x^2 \sqrt{1-(\frac{1}{x})^2}}$ $= \frac{-1}{x^2 \sqrt{\frac{x^2-1}{x^2}}}$ $= \frac{-1}{\frac{x^2}{x} \sqrt{x^2-1}}$ $= \frac{-1}{x \sqrt{x^2-1}}$	(1)	

MATHEMATICS EXTENSION I – QUESTION 13

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$\text{Similarly } \frac{d}{dx} \left(\cos^{-1} \frac{1}{x} \right) \\ = - \left(\frac{-1}{x^2 \sqrt{x^2 - 1}} \right)$$

$$= \frac{1}{x \sqrt{x^2 - 1}}$$

$$\therefore f(x) = \cos^{-1} \left(\frac{1}{x} \right) + \sin^{-1} \left(\frac{1}{x} \right)$$

$$f'(x) = \frac{1}{x \sqrt{x^2 - 1}} + \frac{1}{x \sqrt{x^2 - 1}}$$

$$f'(x) = 0$$

(1)

(iii) Since the gradient is always zero for $x \geq 1$ (ie $f'(x) = 0$ for $x \geq 1$) for $f(x)$, then $f(x)$ must be a horizontal line for $x \geq 1$ (ie $f(x)$ must be a constant in the domain $x \geq 1$)

(1)

$\therefore$ You can sub any x -value, $x \geq 1$ to find $f(x)$.

$$\begin{aligned} f(1) &= \sec^{-1}(1) + \sin^{-1} \left(\frac{1}{1} \right) \\ &= \cos^{-1} \left(\frac{1}{1} \right) + \sin^{-1}(1) \\ &= \cos^{-1}(1) + \sin^{-1}(1) \\ &= 0 + \frac{\pi}{2} \\ &= \frac{\pi}{2} \end{aligned}$$

(1)

$$\begin{aligned} \text{Hence } f(x) &= \sec^{-1} \frac{1}{x} + \sin^{-1} \frac{1}{x} \\ &= \frac{\pi}{2} \end{aligned}$$

MATHEMATICS EXTENSION I – QUESTION 13

SUGGESTED SOLUTIONS

MARKS

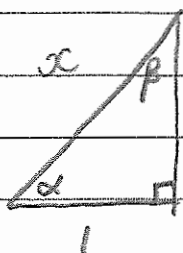
MARKER'S COMMENTS

or $f(x) = \sec^{-1}x + \sin^{-1}\frac{1}{x}$

$$f(x) = \cos^{-1}\frac{1}{x} + \sin^{-1}\frac{1}{x}$$

let $\alpha = \cos^{-1}\frac{1}{x}$
 $\cos \alpha = \frac{1}{x}$

$\beta = \sin^{-1}\frac{1}{x}$
 $\sin \beta = \frac{1}{x}$



$\therefore \alpha + \beta + \frac{\pi}{2} = \pi$ (angle sum right-angled triangle)

$$\alpha + \beta = \pi - \frac{\pi}{2}$$

$$\alpha + \beta = \frac{\pi}{2}$$

①

Using one triangle to represent the information.

①

Using $\angle$ sum of a triangle to prove identity.

or $f(x) = \sec^{-1}x + \sin^{-1}\frac{1}{x}$

$$f(x) = \cos^{-1}\frac{1}{x} + \sin^{-1}\frac{1}{x}$$

To prove $\cos^{-1}\frac{1}{x} + \sin^{-1}\frac{1}{x} = \frac{\pi}{2}$

$$\sin(\cos^{-1}\frac{1}{x} + \sin^{-1}\frac{1}{x}) = \sin\frac{\pi}{2} = 1$$

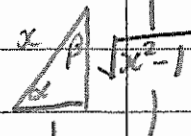
LHS = $\sin[\cos^{-1}\frac{1}{x} + \sin^{-1}\frac{1}{x}]$

let $\alpha = \cos^{-1}\frac{1}{x}$

$\cos \alpha = \frac{1}{x}$

$\beta = \sin^{-1}\frac{1}{x}$

$\sin \beta = \frac{1}{x}$



$$= \sin(\alpha + \beta)$$

$$= \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$= \frac{\sqrt{x^2-1}}{x} \times \frac{\sqrt{x^2-1}}{x} + \frac{1}{x} \times \frac{1}{x}$$

$$= \frac{x^2-1}{x^2} + \frac{1}{x^2}$$

$$= \frac{x^2}{x^2}$$

$$= 1$$

= RHS.

and similarly using cos or tan

①

Using sin correctly

①

correct application

MATHEMATICS EXTENSION I – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
b) $x^2 = 4ay$ $y = px$ Solve simultaneously $x^2 = 4a(px)$ $x^2 = 4apx$ $x^2 - 4apx = 0$ $x(x - 4ap) = 0$ $x = 0$ or $x = 4ap$ sub $x = 4ap$ into $y = px$ $y = p(4ap)$ $y = 4ap^2$ $O(0,0)$ $\therefore C(4ap, 4ap^2)$ Midpoint $x = \frac{0 + 4ap}{2}$ $x = 2ap$ $y = \frac{0 + 4ap^2}{2}$ $y = 2ap^2$ $\therefore D(2ap, 2ap^2)$	$\frac{1}{2}$ $\frac{1}{2}$	Make sure you do not $\div$ by x. (not penalised) Show substitution (not penalised)
(ii) $x^2 = 4ay$ $y = \frac{1}{4a}x^2$ $\frac{dy}{dx} = \frac{2}{4a}x$ $= \frac{x}{2a}$ at $x = 2ap$ $m_T = \frac{2ap}{2a}$ $= p$ <u>OR</u> $x = 2ap$ $y = ap^2$ $\frac{dx}{dp} = 2a$ $\frac{dy}{dp} = ap$ $\frac{dy}{dx} = \frac{dy}{dp} \times \frac{dp}{dx}$ $= ap \times \frac{1}{2a}$ $= p$ $y - ap^2 = p(x - 2ap)$ $y = px - 2ap^2$ when $x = 0$ $y = p(0) - 2ap^2$ $y = -2ap^2$ $B(0, -2ap^2)$	$\frac{1}{2}$ $\frac{1}{2}$	

MATHEMATICS EXTENSION I – QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(iii) $m_{OB} = m_{OA}$ (O and B share the same x-ordinate $\therefore$ vertical line) (D and A share x-ordinate) $\therefore$ vertical	①	One pair sides = and //
$d_{OB} = \sqrt{(-ap^2)^2 + (0-0)^2} = ap^2$ $d_{OA} = \sqrt{(2ap^2-0)^2 + (0-0)^2} = 2ap^2$ $\therefore d_{OB} = d_{OA}$	①	
$\therefore$ ODAB //gm as one pair of sides is both = and //		
OR $m_{OD} = \frac{2ap^2-0}{2ap} = p = m_{AB}$ ODAB //gm as both pairs opp sides //	②	Both pairs sides // all sides =
$d_{DA} = \sqrt{(2ap-2ap)^2 + (ap^2-2ap^2)^2} = \sqrt{a^2p^4} = ap^2$	①	
$d_{OB} = \sqrt{(0-0)^2 + (0+ap^2)^2} = \sqrt{a^2p^4} = ap^2$		
$d_{OD} = \sqrt{(2ap-0)^2 + (2ap^2-0)^2} = \sqrt{4a^2p^2 + 4a^2p^4} = 2ap\sqrt{1+p^2}$ $d_{AB} = 2ap\sqrt{1+p^2}$	①	
ODAB //gm as opp sides =		

MATHEMATICS EXTENSION I – QUESTION 13

[illegible]

MATHEMATICS EXTENSION I – QUESTION 14

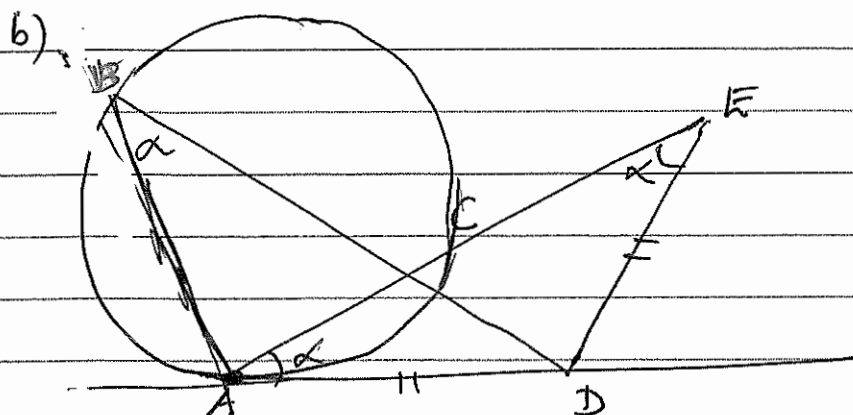
SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$a) f(1.33) = 3\sin(2 \times 1.33) - 1.33$ $= 0.0596 \text{ (4 dp)}$ $f(1.34) = 3\sin(2 \times 1.34) - 1.34$ $= -0.0039 \text{ (4 dp)}$	1/2	Quite well attempted. However, some
	1/2	students failed to
Since $f(x)$ is a continuous function	1/2	leave their
and $f(1.33) > 0$ and $f(1.34) < 0$,	1/2	calculator in
then a root, α , exists such that		radian mode.
$1.33 < \alpha < 1.34$		The majority of students
		lost 1/2 mark
		for not writing
		down that the
		function is
		continuous
$ii) x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \quad f(x) = 3\sin 2x - x$ $f'(x) = 6\cos 2x - 1$	1/2	A few
		students
$= 1.33 - \frac{3\sin(2 \times 1.33) - 1.33}{6\cos(2 \times 1.33) - 1}$	1	need to
		re-visit
$= 1.3394 \text{ (4 dp)}$	1/2	differentiation of
Students need to realise that when solving problems involving the Bisection Method and Newton's Method with respect to trigonometric functions, the calculators should be placed in radian mode.		trigonometric functions
(Degree mode – answer would have been 1.568462878)		
which is not correct.		

MATHEMATICS EXTENSION I – QUESTION 14

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS



$$AD = ED$$

$$\therefore \angle DEA = \angle DAE$$

(equal angles opposite equal sides of an isosceles triangle (ADE))

$\angle DAC = \angle CBA$ (angle between a tangent and a chord through the point of contact is equal to the angle in the alternate segment)

$$\therefore \angle ABD = \angle AED$$

$\therefore ABED$ is a cyclic quadrilateral

(angles in the same segment are equal)

or angles at the circumference standing on the same arc are equal).

Some students wrote down reasons

such as

" $\angle CAD = \angle DBE = \alpha$ (angles subtended by the same arc ED)

This reasoning is based on

the assumption that $ABED$ is

a cyclic quadrilateral. Students were

required to prove that $ABED$ is a

cyclic quadrilateral.

Note: Spelling of isosceles triangle.

Angles had to be

identified and labelled correctly

together with the reasoning otherwise

lack of understanding of the Circle Geometry Theorems is displayed.

MATHEMATICS EXTENSION I – QUESTION 14

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c) $4^n + 6n - 1$ is divisible by 9		
That is, $4^n + 6n - 1 = 9M$		
<u>Step 1</u> Prove true for $n = 1$		
LHS = $4^n + 6n - 1$		
= $4^1 + 6(1) - 1$		
= 9 which is divisible by 9.		
$\therefore$ true for $n = 1$		
<u>Step 2</u> Assume true for $n = k$	1mk	1mk for Step 1, 2 and the conclusion.
That is, $4^k + 6k - 1 = 9Q$,		
Where Q is any integer		
$4^k = 9Q - 6k + 1$		Students should be encouraged to write where Q is any integer (not restricted to positive)
<u>Step 3</u> Prove true for $n = k + 1$		
Prove $4^{k+1} + 6(k+1) - 1$		
= $4^k(4) + 6k + 6 - 1$		
= $4(9Q - 6k + 1) + 6k + 6 - 1$	1	Substituting the assumption
= $36Q - 24k + 4 + 6k + 6 - 1$		
= $36Q - 18k + 9$		
= $9(4Q - 2k + 1)$	1	appropriately into step 3
= $9P$, where $P = 4Q - 2k + 1$		
$\therefore$ divisible by 9.		Appropriate simplification
<u>Conclusion</u>		
It is true for $n = k + 1$ if it is true for $n = k$. Since it is true for $n = 1$, then it is true for $n = 2$, and hence it is true for $n = 3$, and so on.		The conclusion was written down very poorly.
Hence, by the Principle of Mathematical Induction it holds true for all $n \geq 1$		

MATHEMATICS EXTENSION 1 TRIAL HSC 2019 – QUESTION 15 (10 marks)

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
General comments The instructions indicate that relevant mathematical reasoning and/or calculations should be included in the responses for Questions 11–14. Candidates are reminded that where a question is worth several marks, full marks may not be awarded for an answer, even if the answer given is correct, if no working is shown. This is because mathematical communication and reasoning are included in the objectives and outcomes assessed by the examination. Candidates are advised to show all their working so that marks can be awarded for some correct steps towards their answer. A simple example is when candidates have to round their answer to a certain degree of accuracy. Candidates should always write their calculator display before rounding their answer. They should only round their answer in the last step of working, not in an earlier step. Markers can then see that candidates have rounded correctly, even if the answer is not correct.		
Areas for students to improve include: • recognising that the solution is a length and needs to be positive. • paying attention to the mark value of the question and using it as a guide to the complexity of solution required. • “Show that” or “Prove” questions: avoiding the omission of too many steps of the proof and communicating clearly about how they went from one step to the next. • avoiding the omission of too many lines in the algebraic manipulation in an attempt to show the given result. In a ‘show’ question it must be clear how one line is obtained from another. • showing appropriate working and not give an unsupported answer.		
(a)(i) A particle is said to move with simple harmonic motion when the acceleration of the particle about a fixed-point is proportional to its displacement but opposite in direction. Hence, when the displacement is positive the acceleration is negative and vice versa. When asked to prove that, or show that a particle moves with simple harmonic motion, you simply show that the particle satisfies the equation $a = \ddot{x} = -n^2x$ where x is the displacement of the particle about a fixed-point O at time t and n is a positive constant ($n > 0$).		

MATHEMATICS EXTENSION 1 TRIAL HSC 2019 – QUESTION 15 (10 marks)

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(a)(i) In this question, students needed to show that $x = 6\sin 2t$ is a solution to the differential equation $\ddot{x} = -n^2x$.		
$x = 6\sin 2t$		
$\dot{x} = \frac{dx}{dt} = 6\cos 2t \times 2$		
$= 12\cos 2t$		
$\ddot{x} = \frac{d^2x}{dt^2} = -12\sin 2t \times 2$		
$= -24\sin 2t$	①	for finding acceleration
$= -4 \times 6\sin 2t$		
$= -4x$	①	for correct form
$= -(2)^2x$		
Since $\ddot{x} = -4x$ is in the form $\ddot{x} = -n^2x$		
where $n=2$, the particle is moving in simple harmonic motion.		
(ii) Period $= \frac{2\pi}{n}$		
$= \frac{2\pi}{2}$		
$= \pi$	①	provides correct solution

MATHEMATICS EXTENSION 1 TRIAL HSC 2019 – QUESTION 15 (10 marks)

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(iii) when $x=3$:		
$6 \sin 2t = 3$		
$\sin 2t = \frac{1}{2}$	① for correct	
$2t = \frac{\pi}{6}, \frac{5\pi}{6}, \dots$	value of t .	
$t = \frac{\pi}{12}$ when it <u>first</u> reaches $x=3$.		
When $t = \frac{\pi}{12}$: $\dot{x} = 12 \cos(2 \times \frac{\pi}{12})$		
$= 12 \cos \frac{\pi}{6}$		
$= 12 \times \frac{\sqrt{3}}{2}$		
$= 6\sqrt{3} \text{ cm/s}$		
$\doteq 10.39230485 \text{ cm/s}$		]- ① for correct velocity.
• Students who used an incorrect answer for t and who demonstrated the relevant skills were not further penalised if they arrived at a correct solution for $\dot{x}$ for their incorrect t -value.		
① mark - CFPA (it pays to show working and substitutions).		
<u>OR</u> Students who found v^2 from integration where awarded full marks as long as their solution was correct.		

MATHEMATICS EXTENSION 1 TRIAL HSC 2019 – QUESTION 15 (10 marks)

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$		
$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -4x$		
$\frac{1}{2} v^2 = -\frac{4x^2}{2} + C$		
when $t=0$, $v = \dot{x} = 12 \cos 2(0)$		
$x=0$ $a=0$		
$= 12$		
$\frac{1}{2} (12)^2 = 0 + C$		
$\therefore C = 72$		
$\frac{1}{2} v^2 = -2x^2 + 72$		
$v^2 = -4x^2 + 144$		
when $x=3$: $v^2 = -4(3)^2 + 144$		
$= 108$		
$v = \pm \sqrt{108}$		
$= \pm 6\sqrt{3}$		
$\therefore v = +6\sqrt{3} \text{ cm/s}$ with no reference		
to why they chose the positive value over the		
negative value were awarded $1\frac{1}{2}$ marks.		
• some students backed up their answer for		
$v = +6\sqrt{3}$ by sketching x & v and showing that		
at $x=3$, $t = \frac{\pi}{12}$ and hence v is positive at this		
time were awarded 2 marks.		

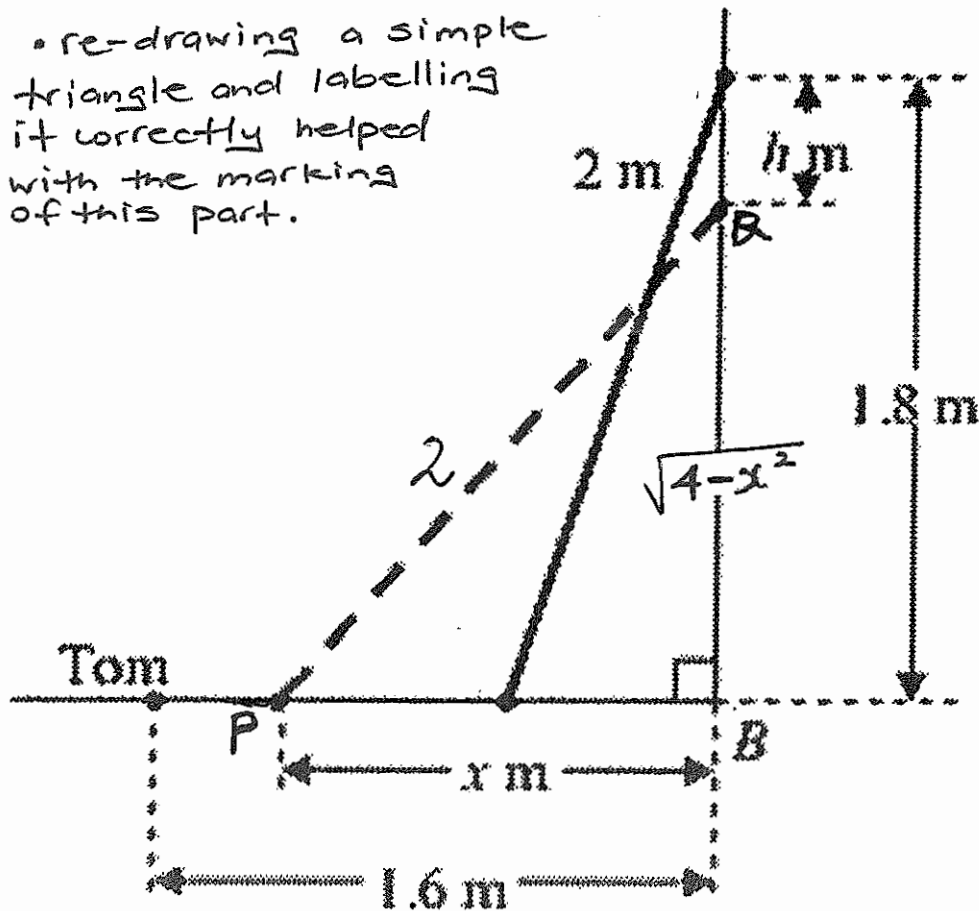
MATHEMATICS EXTENSION 1 TRIAL HSC 2019 – QUESTION 15 (10 marks)

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

(b)(i) • re-drawing a simple triangle and labelling it correctly helped with the marking of this part.



From ΔPAB : Using Pythagoras' Theorem:

$$BQ = \sqrt{PQ^2 - PB^2}$$

$$BQ = \sqrt{2^2 - x^2} \\ = \sqrt{4 - x^2}$$

$$BQ + h = 1.8$$

$$h = 1.8 - BQ$$

$$\therefore h = 1.8 - \sqrt{4 - x^2}$$

① • students needed to show their use of pythagoras theorem for BQ.

① • needed to show statement that $BQ + h = 1.8$ OR $h = 1.8 - BQ$ and show their substitution for BQ.

• Students who used different variables and backed this up with a clear diagram were still awarded 2 marks as long as all steps were shown.

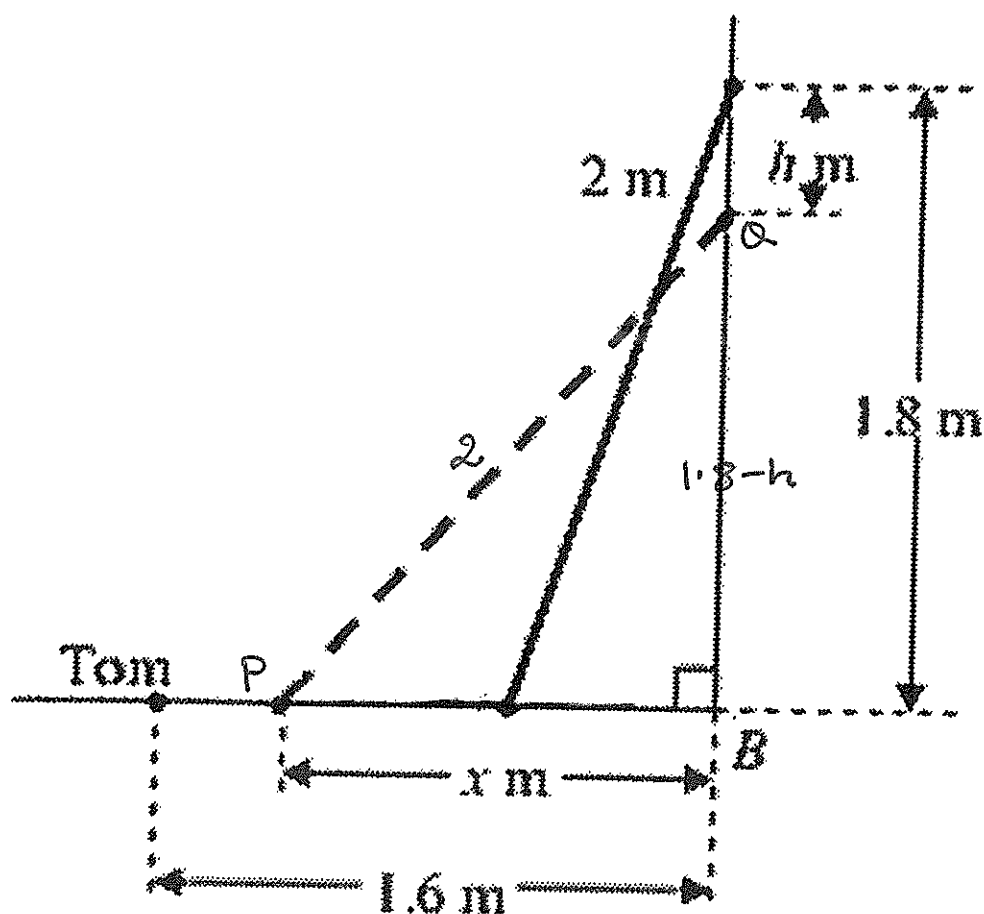
MATHEMATICS EXTENSION 1 TRIAL HSC 2019 – QUESTION 15 (10 marks)

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

OR



BQ

$$= 1.8 - h$$

Using Pythagoras' Theorem for $\triangle PAB$:

$$PQ^2 = BQ^2 + PB^2$$

$$2^2 = (1.8 - h)^2 + x^2$$

$$4 - x^2 = (1.8 - h)^2$$

$$\sqrt{4 - x^2} = 1.8 - h$$

(since $x > 0$)

$$\sqrt{4 - x^2} + h = 1.8$$

$$\therefore h = 1.8 - \sqrt{4 - x^2}$$

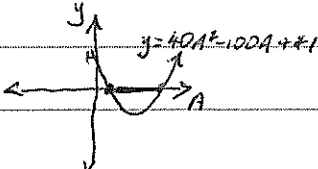
② provides correct solution

- majority of the students used this method and were more successful with their setting out.

MATHEMATICS EXTENSION 1 TRIAL HSC 2019 – QUESTION 15 (10 marks)

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(b)(ii) using (i)		• many errors made
$h = 1.8 - \sqrt{4-x^2}$		when differentiating h w.r.t. x .
$h = 1.8 - (4-x^2)^{\frac{1}{2}}$		• (some integrated and still wrote their answer as $\frac{dh}{dx}$)
$\frac{dh}{dx} = -\frac{1}{2}(4-x^2)^{-\frac{1}{2}} \times -2x$		(-1) for this.
$\frac{dh}{dx} = \frac{x}{\sqrt{4-x^2}}$	① for correct $\frac{dh}{dx}$	
$\therefore \frac{dx}{dh} = \frac{\sqrt{4-x^2}}{x}$	NOTE:	$\frac{dh}{dt} = 0.5$ m/min
$\frac{dh}{dt} = 0.5$ m/min		not 1.6.
Using chain rule: $\frac{dx}{dt} = \frac{dx}{dh} \times \frac{dh}{dt}$		• showing all working with substitutions made marking this question easier as I could check answers using my calculator CFPA.
$= \frac{\sqrt{4-x^2}}{x} \times 0.5$		• Bald answers were not awarded any marks!
When $x=1.6$:		
$\frac{dx}{dt} = \frac{\sqrt{4-1.6^2}}{1.6} \times 0.5$	① or equivalent merit	
$= \frac{\sqrt{1.44}}{1.6} \times 0.5$		• many students forgot to reciprocate their $\frac{dh}{dx}$ (=1) leading to an answer of $\frac{2}{3}$ m/min (2 marks awarded).
$= \frac{1.2}{1.6} \times 0.5$		
$= 0.75 \times 0.5$		
$= 0.375$ m/min	① for correct answer	
$= \frac{3}{8}$ m/min		
• students who used $\frac{dh}{dt} = -0.5$ and showed all working, arrived at -0.375 m/min were awarded (2½) marks.		

MATHEMATICS EXTENSION 1 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) (i) $50t = x \sec \theta$ $t = \frac{x \sec \theta}{50}$ $y = 50 \frac{x \sec \theta}{50} \sin \theta - 5 \left(\frac{x \sec \theta}{50} \right)^2$ $= x \frac{\sin \theta}{\cos \theta} - \frac{5x^2 \sec^2 \theta}{2500}$ $= x \tan \theta - \frac{x^2}{500} \sec^2 \theta$	1 1	1 mark for making t the subject and 1 mark for correct substitution into y
(ii) When $x = 200$ $y = 2$ $2 = 200 \tan \theta - \frac{200^2}{500} \sec^2 \theta$ $= 200 \tan \theta - 80 \sec^2 \theta$ $= 200 \tan \theta - 80 (\tan^2 \theta + 1)$ $= 200 \tan \theta - 80 \tan^2 \theta - 80$ $82 = 200 \tan \theta - 80 \tan^2 \theta$ $41 = 100 \tan \theta - 40 \tan^2 \theta$ $\Rightarrow 40 \tan^2 \theta - 100 \tan \theta + 41 = 0$	1 1	1 mark for substitution 1 mark for using $\sec^2 \theta = \tan^2 \theta + 1$
(iii) Let $A = \tan \theta$ $40A^2 - 100A + 41 = 0$  $A = \frac{100 \pm \sqrt{100^2 - 4 \times 40 \times 41}}{80}$ $= \frac{100 \pm \sqrt{3440}}{80}$ $= \frac{25 \pm \sqrt{215}}{20}$ $\frac{25 - \sqrt{215}}{20} \leq A \leq \frac{25 + \sqrt{215}}{20}$ $\frac{25 - \sqrt{215}}{20} \leq \tan \theta \leq \frac{25 + \sqrt{215}}{20}$ $27^\circ \leq \theta \leq 63^\circ \quad \text{to the nearest degree}$	1	

MATHEMATICS EXTENSION 1 – QUESTION 16

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
b)(i) Area of rectangle $1 \times k = k$ Area enclosed $= k - \int_0^k \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$ If $f(x) = e^x + e^{-x}$ $f'(x) = e^x - e^{-x}$ using $\int_0^k \frac{f'(x)}{f(x)} dx = [\ln f(x)]_0^k$ Area enclosed $= k - [\ln e^x + e^{-x}]_0^k$ $e^k + e^{-k} > 0$ $= k - (\ln(e^k + e^{-k}) - \ln(e^0 + e^{-0}))$ $= k - (\ln(e^k + e^{-k}) - \ln 2)$ $= k - \ln(e^k + e^{-k}) + \ln 2$	1	1 mark for setting up the integral
(ii) Since $(e^k + e^{-k}) > e^k$ ($e^{-k} > 0$) $\ln(e^k + e^{-k}) > \ln e^k$ $-\ln(e^k + e^{-k}) < -\ln e^k$ $k - \ln(e^k + e^{-k}) < k - \ln e^k$ $k - \ln(e^k + e^{-k}) < k - k$ $k - \ln(e^k + e^{-k}) < 0$ $k - \ln(e^k + e^{-k}) + \ln 2 < \ln 2$ $\therefore A < \ln 2$ for all $k > 0$	1	1 mark for finding the integral
(ii) Since $(e^k + e^{-k}) > e^k$ ($e^{-k} > 0$) $\ln(e^k + e^{-k}) > \ln e^k$ $-\ln(e^k + e^{-k}) < -\ln e^k$ $k - \ln(e^k + e^{-k}) < k - \ln e^k$ $k - \ln(e^k + e^{-k}) < k - k$ $k - \ln(e^k + e^{-k}) < 0$ $k - \ln(e^k + e^{-k}) + \ln 2 < \ln 2$ $\therefore A < \ln 2$ for all $k > 0$	1	1 mark for setting this up
(ii) Since $(e^k + e^{-k}) > e^k$ ($e^{-k} > 0$) $\ln(e^k + e^{-k}) > \ln e^k$ $-\ln(e^k + e^{-k}) < -\ln e^k$ $k - \ln(e^k + e^{-k}) < k - \ln e^k$ $k - \ln(e^k + e^{-k}) < k - k$ $k - \ln(e^k + e^{-k}) < 0$ $k - \ln(e^k + e^{-k}) + \ln 2 < \ln 2$ $\therefore A < \ln 2$ for all $k > 0$	1	1 mark for completing the proof
All sections of Q16 were done well except for b)(ii) which was poorly done		